

Aaron Randolph Arada

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EDUCATION

FEU Institute of Technology

Bachelor of Science in Computer Science with specialization in Software Engineering

Manila, Philippines

Aug. 2024 - Present

PROJECTS

KUMPAS (FSL Translator) | *React, Python, TensorFlow, MediaPipe*

June 2026 | [Live Demo](#)

- Built during ACM TechSprint: Asteria, a machine learning-powered Filipino Sign Language recognition system supporting **29 gesture classes**, using webcam input and MediaPipe hand landmarks to detect FSL gestures in real time
- Trained an LSTM sequence classification model with TensorFlow/Keras on **30-frame gesture sequences** containing **126 hand-landmark features per frame**
- Deployed a full-stack translator with a Vercel frontend and Railway Python inference backend, supporting translation output across **6 Philippine languages**

Aaron Intelligence (AI Portfolio Assistant) | *React, FastAPI, Supabase, Gemini API*

June 2026 | [Live Demo](#)

- Built a RAG-based portfolio assistant over **30+ knowledge documents**, enabling users to query projects, skills, education, and experience through natural-language conversations
- Implemented semantic search using Sentence Transformers, pgvector, and Supabase PostgreSQL, creating a scalable vector retrieval pipeline for context-aware AI responses
- Developed and deployed a full-stack application using React, FastAPI, and Google Gemini API, providing an interactive AI-powered alternative to traditional portfolio navigation

Mirror Mentor (Hackathon Project Lead) | *FastAPI, Supabase, Gemini API, Vercel*

May 2026

- Led backend development for an AI-powered educational platform during ACM Developers Week, designing and maintaining **17 REST API endpoints** supporting authentication, conversations, and AI-assisted code analysis
- Integrated Google Gemini API with a **4-phase Socratic questioning framework** to deliver guided programming feedback through multi-step code analysis workflows
- Implemented secure authentication, real-time data synchronization, and persistent conversation storage using Supabase Auth, RLS, Realtime, and PostgreSQL while validating usability with **5 student users**

AlgoVisualizer | *React, TypeScript, TailwindCSS*

June 2026

- Built an interactive algorithm visualization platform that demonstrates sorting and searching algorithms through real-time animations and step-by-step execution
- Implemented configurable controls for execution speed, array generation, and algorithm selection, enabling users to explore algorithm behavior and performance interactively
- Developed educational visualizations and complexity references to improve understanding of fundamental data structures and algorithms concepts

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C++

Frontend: React, Next.js, TailwindCSS, shadcn/ui, HTML/CSS

Backend: FastAPI, Node.js, REST APIs, Supabase, PostgreSQL

AI/ML: TensorFlow/Keras, LSTM, RAG, Gemini API, Sentence Transformers, OpenCV, MediaPipe

Cloud/Tools: AWS EC2, AWS S3, Vercel, Railway, Git, DBBeaver, VS Code

CERTIFICATIONS

AWS Academy Generative AI Foundations – AWS Academy, Expected 2026

Pearson IT Specialist: Python – Certiport, Expected Jul. 2026

Microsoft Office Specialist: Excel Associate (Microsoft 365 Apps) – Microsoft, Aug. 2023